

# Transect / Cross-section

Draw a line across the basin and see the water beneath it - depth against distance, with contours and real floats on top.

| PORT | PAGE FILE                   | MODULE                     | SAMPLING                 |
|------|-----------------------------|----------------------------|--------------------------|
| 8510 | app/phase2/transect_page.py | phase2/derived/transect.py | bilinear, gaps preserved |

This is how an oceanographer actually reads structure. A tilting thermocline, an eddy's subsurface core, a coastal upwelling gradient - all of them are obvious in a vertical slice and nearly invisible in a map or a single profile.

## 1. What this feature is, in plain words

You give it two points. It cuts a vertical wall through the ocean between them and colours it by temperature: distance along the bottom, depth down the side. Everything the model knows along that line appears at once.

Two honesty rules are enforced in the mathematics rather than in a caption. First, the slice is sampled by **bilinear interpolation**, because the point predictor snaps to the nearest grid centre and sampling it directly would draw a staircase that looks like real structure. Second, land and the sea floor are **gaps**, never values smoothed across - an interpolated cell touching a missing corner is itself missing. So a blank in the section is the coast or the bottom, never a hole in the model output.

The upgraded version puts the model and the reanalysis side by side, scatters real independent floats on top coloured by the model's error against them, and lets you slide the whole track east or west to watch a front move.

### THE EVERYDAY PICTURE

A map is a photograph of the sea's skin. A transect is the cut through a layer cake - and the layers are the entire point.

## 2. What you actually see on the screen

One line, several ways of checking it:

- The model's temperature section - distance along the track against depth.
- The GLORYS reanalysis section beside it, on the same colour scale.
- Real independent Argo floats scattered on the section, coloured by model-minus-float error, each carrying its distance in kilometres and its time offset in days into the tooltip.
- **Isopycnal** (constant density) and **sound-speed** contours, where the chosen source carries salinity.
- A sliding control that sweeps the same track sideways, preserving its length exactly.
- A date picker over the model's real calendar - not a free-text box.

## 3. How to run it, step by step

### RUN THIS

```
PYTHONPATH=src python -m streamlit run app/phase2/transect_page.py --server.port 8510
```

- 1 Run the command and open **http://localhost:8510**.
- 2 Enter two endpoints. A good demonstration track runs 8 N, 68 E to 20 N, 88 E - it crosses the Arabian Sea, the tip of India and into the Bay of Bengal.
- 3 Pick a date from the calendar picker.
- 4 Read the model section, then reveal the GLORYS section beside it. State clearly that GLORYS is the training target, so agreement there is agreement with what we were fitted to - not independent proof.
- 5 Turn on the Argo scatter. **These** are the independent checks, and each one shows how far away it was.
- 6 Use the slider to sweep the track and let the audience watch the structure move. Point out that the track length never changes as it slides.

### THREE REAL BUGS THIS PAGE FIXED, WORTH MENTIONING

Its cache key was never actually passed, so the page would not refresh when the model changed - the exact mechanism behind an 8 degC dashboard error earlier in the project. Its date box accepted 1850-01-01 and silently returned a plausible section. And its legend *understated* the model's uncertainty by saying the band was uncalibrated when it was calibrated - the only honesty bug in this project that claimed less rather than more.

## 4. Why the problem statement needs this

The problem statement asks for a depth-resolved product. A transect is the standard scientific way to present one, and it is the view an oceanographer on a review panel will look for first.

It is also where model and reanalysis can be compared like for like, with independent floats scattered on top as the tie-breaker - three sources on one picture, each labelled for what it is.

| PS requirement                       | How this feature answers it                                                                                 |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Depth-resolved reconstruction</b> | Renders the full 0-1000 m column continuously along any track the user draws.                               |
| <b>Independent validation</b>        | Floats are overlaid with their real distance and time offsets, so a weak check cannot pass as a strong one. |
| <b>0.25 degree resolution</b>        | Bilinear sampling of the frozen grid, with land and sea floor preserved as gaps.                            |

## 5. The measured numbers behind it

Measured on the default track [VERIFIED]:

| Comparison                                     | Result                                                                                                                | How to read it                                                                                                                                                           |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Model against GLORYS along the section</b>  | bias +0.011 degC, RMSE 0.560 degC over 747 section points                                                             | GLORYS is the <b>training target</b> , so this is agreement with what we were fitted to. The caption says so, which stops the smaller number standing as the better one. |
| <b>Model against 3 independent Argo floats</b> | pooled RMSE 0.795 degC, float offsets 2-60 km                                                                         | <b>This</b> is the independent check - and the 60 km float is a weaker one than the 2 km float, which is why every offset is shown.                                      |
| <b>Isopycnal contour resolution</b>            | 24 kg/m3 surface resolved at <b>26 of 60</b> points by the density contour, and <b>0 of 60</b> by the isotherm method | The shared-foundation work paying for itself, now pinned by a test.                                                                                                      |

- Asking the sampler for a quantity the field does not carry **raises an error** rather than returning an all-missing section, because an all-missing density overlay is indistinguishable from 'there are no isopycnals here'.

- The sliding control clips the **shift** rather than each endpoint independently. Clipping the endpoints separately lets one end stop at the boundary while the other keeps moving, so the track silently shortens - a control that promises to hold a line's shape and quietly deforms it. The span is now preserved by construction.

## 6. The impact - and who it is for

| Who                                | What changes for them                                                                                                          |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Ocean scientists</b>            | The standard diagnostic view, on a daily satellite-driven product - including features a monthly average would erase entirely. |
| <b>Naval and acoustic planners</b> | Sound-speed structure along a chosen route, which is exactly how an acoustic path is assessed.                                 |
| <b>The jury</b>                    | Three sources on one picture, each labelled for what it is, with the weakest check shown rather than hidden.                   |
| <b>Educators</b>                   | The clearest teaching image for ocean layering, built from real reconstruction rather than a textbook schematic.               |

### WHY SOCIETY GAINS FROM THIS

Structure, not average temperature, is what determines where fish gather, how far sound travels and how a storm draws energy. A tool that renders that structure daily along any line, from freely available satellite data, lowers the cost of asking those questions from a research cruise to a browser tab.

## 7. Say this to the jury

*Draw any line across the basin and this cuts a vertical wall through the ocean beneath it. The model is on the left, the reanalysis on the right, and real independent floats are scattered on top coloured by our error against them. Note what the captions say: against GLORYS we score 0.560 degC, but GLORYS is our training target, so that is agreement with what we were fitted to. Against the independent floats we score 0.795, and each float shows how far away it was - because a float 60 km off is a weaker check than one 2 km off, and hiding that would overstate our validation.*

## 8. Questions the jury will ask

### Q. Why is your error against GLORYS smaller than against floats?

**A.** Because GLORYS is what we were trained to reproduce. Agreement with your own training target is not independent evidence, and the caption on the page says exactly that. The 0.795 degC against real floats is the number that means something.

### Q. Are those blank areas model failures?

**A.** No - they are land and the sea floor. Interpolation here is designed so that any cell touching a missing corner is itself missing, so a blank is always the coast or the bottom, never a gap in our output.

### Q. Why bilinear sampling instead of just reading the grid?

**A.** Because the point predictor snaps to the nearest grid centre, and sampling it directly along a diagonal track would draw a staircase that a viewer would read as real structure. Bilinear sampling of the field is the honest rendering of what the model actually says.

## 9. The honest limit - say it before they find it

### IT INHERITS THE GRID'S RESOLUTION

At 0.25 degrees - roughly 25 km - sub-mesoscale structure simply cannot be resolved, and a smooth section should not be read as evidence that the ocean is smooth. The density and sound-speed contours also need salinity, so they are only available on sources that carry it, and the page names which source is active rather than blending them.

#### IN ONE LINE

*A vertical wall cut through the ocean along any line, with the model, the reanalysis and independent floats on one picture and each labelled for what it is.*

**0.560 degC**

VS GLORYS - OUR TRAINING TARGET

**0.795 degC**

VS INDEPENDENT FLOATS - THE REAL CHECK

**26 of 60**

ISOPYCNAL POINTS THE DENSITY CONTOUR  
RESOLVES

Feature 10 of 17. Next: Feature 11.